Ego-Conditioned Pedestrian Body Motion Generation via Latent Diffusion with Full-Sequence Ego Conditioning

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Abstract

We introduce EgoPed, a system that generates realistic 3D full-body pedestrian motions conditioned on vehicle ego trajectory observations from autonomous driving data. Existing pedestrian motion synthesis methods rely on text descriptions or action class labels, ignoring the driving context that fundamentally shapes pedestrian behavior — how a person walks, stops, or reacts is strongly influenced by nearby vehicle speed and proximity. We condition a Motion Latent Diffusion (MLD) model on vehicle ego odometry by exposing the denoiser to the full 196-timestep ego trajectory sequence (rather than a pooled summary), using a frozen ego encoder that produces per-timestep conditioning tokens available to the denoiser at every diffusion step. Experiments on a combined dataset built from nuScenes, Waymo, and recordings from our sensor-equipped research vehicle (AVA) demonstrate a **49% FID improvement** ($6.603 \to 3.392$) over the strongest pooled-conditioning baseline, and an ablation study reveals that the frozen encoder is critical: unfreezing causes a 34% MultiModality collapse ($3.956 \to 2.618$) as the encoder over-specializes for discriminative per-condition features. Our best system achieves FID $= 3.392 \pm 0.179$ with R-Precision@$1 = 0.548 \pm 0.002$ at CFG $= 10$, providing a strong foundation for generating behavior-consistent pedestrian motions for AV simulation and safety-critical scenario generation.

1 Introduction

Generating realistic pedestrian body motions is a critical capability for autonomous vehicle (AV) simulation. Simulation pipelines require diverse, plausible pedestrian behaviors — walking, turning, stopping, reacting to traffic — to stress-test perception and planning systems before deployment. While substantial progress has been made in text-conditioned 3D human motion generation Chen et al. [2023], Tevet et al. [2023], Zhang et al. [2024], Guo et al. [2024], these systems generate motions that are context-agnostic: a pedestrian crossing a street looks identical regardless of whether a vehicle is approaching at 50 km/h or stationary.

In reality, pedestrian behavior is strongly coupled to the surrounding vehicle context. The ego vehicle's speed, trajectory, and proximity all influence whether a pedestrian will cross, wait, accelerate, or turn. This *ego-conditioned* setting — generating full 3D body motion conditioned on vehicle ego trajectory — is largely unexplored. Existing work in this space either generates 2D trajectory distributions only Mao et al. [2023], or uses ego context as scene conditioning without 3D body synthesis Feng et al. [2024].

This paper addresses ego-conditioned *full-body* pedestrian motion synthesis. We build on Motion Latent Diffusion (MLD) Chen et al. [2023], which compresses 263D HumanML3D body motion into a compact latent space and denoises within it, but replace the text conditioning with vehicle ego trajectory. The key architectural question is: *how should the ego trajectory be injected into the denoiser?*

We study two families of conditioning mechanisms:

- **Pooled conditioning (H2)**: The ego trajectory is encoded by a transformer encoder and mean-pooled to a single 256D token, which is concatenated alongside the time embedding in the denoiser's self-attention. This gives the denoiser a coarse summary of the ego trajectory but no temporal resolution.
- **Full-sequence conditioning (H4)**: The ego trajectory is encoded to a full 196-token sequence, which is concatenated with the latent tokens and jointly self-attended at each denoising step. This exposes every denoising step to the full ego trajectory, enabling temporally fine-grained conditioning.

Our main finding is that full-sequence conditioning dramatically outperforms pooled conditioning. H4 achieves **FID = 3.392**, a 49% improvement over the best H2 result (FID = 6.603), with the same denoiser capacity — and its best checkpoint occurs *earlier* in training (epoch 3399 vs. 4399), so the improvement does not come from additional compute. The full-sequence denoiser can attend to specific moments in the ego trajectory rather than a coarse average, producing motion that more closely follows the ground truth distribution.

A secondary but equally important finding concerns the ego encoder's training regime. H4 uses a *frozen* ego encoder — pretrained to align mean-pooled ego trajectories with VAE motion latents, then fixed during diffusion training. We ablate this choice with H6, which unfreezes the encoder and allows it to co-adapt with the denoiser. H6 consistently *underperforms* H4 across all evaluated checkpoints: FID worsens by 50%, R-Precision@1 drops by 36%, and MultiModality collapses by 34%. We show that this occurs because the unfrozen encoder specializes to discriminative, per-condition representations that over-constrain generation diversity — consistent with the frozen-encoder paradigm in image generation (frozen CLIP in Stable Diffusion Rombach et al. [2022], adapters on frozen backbones in IP-Adapter Ye et al. [2023] and ControlNet Zhang et al. [2023b]).

Our contributions are:

1. The first system to generate full 3D HumanML3D body motions conditioned on vehicle ego odometry, trained on AVA, nuScenes, and Waymo data.
2. A demonstration that full-sequence temporal conditioning achieves 49% FID improvement over single-token pooled conditioning, with non-overlapping confidence intervals across 3 evaluation replications; an ablation further shows this holds regardless of whether the ego sequence is injected by self-attention over the concatenated sequence or by cross-attention.
3. An ablation showing that frozen ego encoders are critical for generative diversity in ego-conditioned motion: unfreezing causes MultiModality collapse and worsens all metrics.
4. A design recommendation: frozen ego encoder + full-sequence conditioning (H4) is the correct architecture for this task.

2 Related Work

Ego-conditioned and context-conditioned motion generation. The closest works to ours generate 2D pedestrian trajectories conditioned on ego-vehicle or scene context. LED Mao et al. [2023] uses latent-space diffusion for stochastic trajectory prediction conditioned on past observations, demonstrating that latent-space diffusion substantially outperforms raw-space approaches for trajectory generation. UniTraj Feng et al. [2024] trains a unified trajectory-prediction model across multiple AV datasets (nuScenes, Argoverse 2, Waymo Open Motion), demonstrating the value of multi-dataset training for agent behavior modeling — though for 2D trajectory prediction rather than 3D body synthesis. Our work differs in generating 263D full-body motion in the HumanML3D representation, enabling realistic body pose, gait, and foot contact — not just root position.

Interaction-aware and retrieval-augmented motion generation. ReMoDiffuse Zhang et al. [2023c] augments motion diffusion with retrieved nearest-neighbour motions cross-attended into the denoiser, an architectural pattern related to ours where external sequences condition the denoiser via cross-attention. InterGen Liang et al. [2024] jointly generates two-person interactions with mutual cross-attention. Our setting is a degenerate version where one agent (the ego vehicle) has a fixed, observed trajectory — enabling a simpler asymmetric design in which the observed ego trajectory conditions the denoiser as a fixed token sequence.

Latent diffusion for human motion. MLD Chen et al. [2023] is our backbone: it diffuses in the latent space of a pre-trained HumanML3D VAE, using a transformer denoiser conditioned on CLIP text embeddings. We retain the VAE, denoiser architecture, evaluation metrics, and DDIM sampling from MLD, replacing text conditioning with ego trajectory. MDM Tevet et al. [2023] operates in raw motion space and conditions on text, action labels, or trajectory waypoints. T2M-GPT Zhang et al. [2023a] uses discrete VQ-VAE tokenization — less suitable for continuous ego trajectory signals. MoMask Guo et al. [2024] achieves state-of-the-art FID on HumanML3D with masked transformer generation; we note that our metrics are computed in the same evaluation protocol, though on a different ego-conditioned dataset split.

Classifier-free guidance. CFG Ho and Salimans [2022] trains conditional and unconditional models jointly, enabling quality-diversity tradeoff at inference. We find that CFG interacts strongly with conditioning granularity: pooled conditioning (H2) shows monotonically increasing FID with CFG scale, while full-sequence conditioning (H4) is nearly CFG-insensitive. This suggests that richer conditioning naturally prevents the over-constraint that degrades FID at high CFG.

Frozen encoders in generative models. A recurring pattern in conditional image generation is that frozen encoders preserve generative diversity better than fine-tuned ones. Stable Diffusion Rombach et al. [2022] freezes CLIP during diffusion training; IP-Adapter Ye et al. [2023] and ControlNet Zhang et al. [2023b] add conditioning adapters on frozen diffusion backbones. Conversely, approaches that fine-tune the generative stack end-to-end, such as DreamBooth Ruiz et al. [2023], are known to trade diversity for condition fidelity and require explicit prior-preservation regularization to counteract it. Our H6 ablation provides direct empirical evidence for the same phenomenon in motion generation: unfreezing the ego encoder degrades generation quality and collapses per-condition diversity.

3 Method

3.1 Problem Formulation

Given a vehicle ego trajectory $\mathbf{e} \in \mathbb{R}^{T \times 2}$ (2D position over $T = 196$ timesteps), we aim to generate a plausible 3D pedestrian body motion $\mathbf{x} \in \mathbb{R}^{T \times 263}$ in the HumanML3D representation Guo et al. [2022]. The 263D representation encodes root velocity, root angular velocity, joint positions, velocities, and foot contact labels, fully parameterizing a SMPL-like body in motion.

We model the conditional distribution $p(\mathbf{x} \mid \mathbf{e})$ using a latent diffusion model that denoises within a pre-trained VAE's latent space.

3.2 Motion VAE

We use the MLD VAE Chen et al. [2023], a Transformer encoder-decoder that compresses $T \times 263$ motion to a latent $\mathbf{z} \in \mathbb{R}^{L \times 256}$ with $L = 4$ latent tokens (L is a hyperparameter; we find $L = 4$ significantly outperforms $L = 8$, see Section 5). The VAE is pre-trained on the HumanML3D dataset and held fixed throughout diffusion training.

3.3 Ego Encoder

The ego trajectory $\mathbf{e} \in \mathbb{R}^{196 \times 2}$ is encoded by a lightweight transformer encoder:

$$\mathbf{c} = \mathrm{EgoEncoder}(\mathbf{e}) \in \mathbb{R}^{196 \times 256}. \tag{1}$$

The encoder consists of a linear projection from 2D to 256D, followed by sinusoidal positional encodings, and $N = 4$ transformer encoder layers with 4 heads. Unlike the pooled variant (H2), which applies mean-pooling to produce a single token $\mathbf{c} \in \mathbb{R}^{1 \times 256}$, our encoder preserves the full temporal sequence.

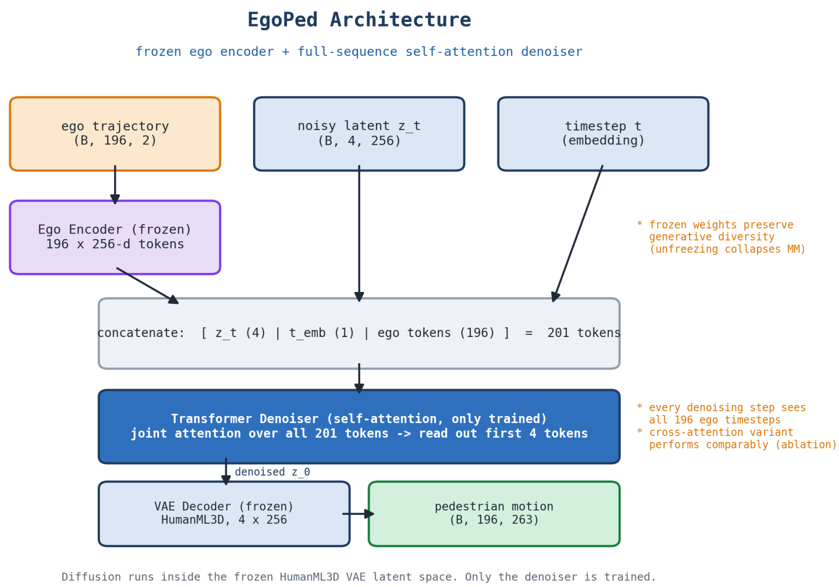

Figure 1: EgoPed architecture. The ego trajectory is encoded by a *frozen* transformer encoder into 196 per-timestep conditioning tokens. At every denoising step, the noisy latent $\mathbf{z}_t$ is concatenated with the full 196-token ego sequence (and the timestep embedding) and jointly processed by a self-attention transformer denoiser; the latent output tokens predict the noise. Diffusion runs in the latent space of a frozen HumanML3D VAE; only the denoiser is trained. Section 5.3 ablates an alternative cross-attention denoiser.

Ego encoder pretraining. Before diffusion training, the ego encoder is pretrained with a regression-based alignment objective that distills the paired motion's VAE latent into the ego encoding. Specifically, we align the mean-pooled ego encoding $\bar{\mathbf{c}} = \mathrm{MeanPool}(\mathbf{c}) \in \mathbb{R}^{256}$ with the mean-pooled VAE posterior mean $\bar{\mathbf{z}} \in \mathbb{R}^{256}$ (computed from the paired motion) using a combined MSE and cosine objective:

$$\mathcal{L}_{\text{pretrain}} = \|\bar{\mathbf{c}} - \bar{\mathbf{z}}\|^2 + \lambda_{\cos}\big(1 - \cos(\bar{\mathbf{c}}, \bar{\mathbf{z}})\big), \tag{2}$$

where the cosine term encourages directional alignment in addition to magnitude. After pretraining, the encoder achieves validation MSE ≈ 1.92, matching the pooled variant (which uses an additional 2-layer projection head without improvement), confirming that the projection head adds no alignment capacity.

Frozen encoder (key design choice). During diffusion training, the ego encoder parameters are *frozen*. This preserves the encoder's learned alignment while ensuring the generated motion distribution is not over-constrained to particular ego conditions. We ablate this choice in Section 5.

3.4 Diffusion Denoiser

We use the MLD denoiser as a self-attention transformer that operates over a single concatenated token sequence. At each denoising step, the $L = 4$ noisy latent tokens $\mathbf{z}_t$, the timestep embedding $\mathbf{t}_{\text{emb}}$, and the full $T = 196$-token ego conditioning sequence $\mathbf{c}$ are concatenated into one sequence of length $L + 1 + T = 201$ and processed by the transformer; the first L output tokens are read out as the predicted noise:

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c}) = \mathrm{Transformer}\big([\mathbf{z}_t,\ \mathbf{t}_{\text{emb}},\ \mathbf{c}]\big)_{1:L}, \tag{3}$$

where the transformer applies self-attention over the joint sequence and the sampler computes $\mathbf{z}_{t-1}$ from $\hat{\boldsymbol{\epsilon}}$. Because every ego token participates in the same self-attention as the latent tokens, each denoising step has access to the full ego trajectory at per-timestep resolution: the denoiser can attend to specific moments in the ego trajectory (e.g., a braking event or sharp turn) rather than a global average. This is the key distinction from the pooled baseline (H2), whose denoiser sees only a single mean-pooled ego token ($L + 2$ tokens in total) and therefore has no temporal resolution over the ego trajectory.

The added cost over the pooled baseline is modest at this scale: self-attention over 201 tokens versus $L + 2$. A natural alternative is a *cross-attention* denoiser, in which the L latent tokens act as queries and the ego sequence as keys/values, reducing the attention cost to $\mathcal{O}(L \times T)$ rather than $\mathcal{O}((L+T)^2)$. We evaluate this variant in Section 5 and find it performs *comparably*, indicating that the benefit comes from exposing the denoiser to the full ego sequence, not from the specific attention mechanism used to inject it.

3.5 Classifier-Free Guidance

We train with classifier-free guidance Ho and Salimans [2022]: for 10% of training samples, the input ego trajectory is zeroed before encoding, so the encoder's response to the all-zero trajectory serves as the unconditional embedding (the same construction is used for the unconditional branch at inference). At inference, the guided score is:

$$\hat{\boldsymbol{\epsilon}}_t = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \emptyset) + w \cdot \big(\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \mathbf{c}) - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \emptyset)\big), \tag{4}$$

with guidance scale w. We find $w = 10$ optimal for H4 (Section 4).

3.6 Training Objective

The denoiser is trained with the standard DDPM noise-prediction objective:

$$\mathcal{L} = \mathbb{E}_{\mathbf{z}_0, \boldsymbol{\epsilon}, t}\big[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c})\|^2\big]. \tag{5}$$

We use DDIM Song et al. [2021] for fast inference with 50 denoising steps.

4 Experiments

4.1 Dataset

As there are no datasets available for the problem investigated in this work, we generated a dataset for ego-conditioned motion generation from both existing autonomous driving datasets and from scenes recorded with our sensor-equipped research vehicle AVA. The vehicle is equipped with seven 128-layer LiDARs and eight RGB cameras for full 360° sensor coverage around the vehicle. Figure ?? shows the vehicle and a detailed sensor setup. To obtain scenes with meaningful interaction between the vehicle and a pedestrian, we staged NUMBER highly interactive scenarios. An example scene of a pedestrian waving a vehicle through can be seen in Figure ??. The recorded data is brought into nuScenes format and objects in the scene are annotated with 3D bounding boxes and orientation labels using MS3D Tsai et al. [2024, 2023].

With the data in nuScenes format fully annotated, we extract 3D human poses in SMPL Loper et al. [2015] format using an adapted version of OmniRe Chen et al. [2025] from all three data sources: AVA, nuScenes, and Waymo Open Dataset. We estimate poses for every pedestrian instance from all available camera images, using the projected 3D bounding box to localize the person in each view. Per-instance pose estimates from different cameras are then stitched into a single continuous sequence using the extrinsic calibration of the camera rigs. Finally, the SMPL pose sequences are converted to the 263D HumanML3D motion representation Guo et al. [2022]. In addition to the pedestrian pose sequences, we extract the synchronized ego-vehicle trajectory from each scene's odometry. A single automated pipeline processes every (scene, pedestrian) pair end-to-end: it loads the annotated nuScenes-format scene, extracts and stitches the per-instance SMPL body sequence, resamples both the body motion and the ego odometry to a common 20 fps grid, and expresses both streams in a *pedestrian-centric* coordinate frame (translation- and yaw-normalized to the pedestrian) so that conditioning is invariant to the absolute world pose of the scene. Each resulting sample is stored as a self-contained record containing the pedestrian's 22 body-joint positions (`ped_in_ped_frame` $\in \mathbb{R}^{T \times 22 \times 3}$), the ego trajectory in the same frame (`ego_in_ped_frame` $\in \mathbb{R}^{T \times 3}$), and the derived 263D HumanML3D motion vector (`vectors_263` $\in \mathbb{R}^{T \times 263}$), where the native sequence length T varies per scene and is windowed to 196 frames at training time (see the interaction-aware cropping procedure below).

Table 1: Dataset composition. Each sample is a paired sequence of ego odometry (196×2) and pedestrian body motion (196×263), extracted from AVA, nuScenes, and Waymo. Train/validation splits are disjoint at the scene level; the validation split is the evaluation set used for all reported metrics.

Source	Train	Validation	Total
AVA (ours)	27	7	34
nuScenes	3,637	943	4,580
Waymo Open Dataset	5,876	1,464	7,340
Total	9,540	2,414	11,954

Using all three data sources we obtain **11,954** paired ego–pedestrian motion samples in total, split by scene into 9,540 training and 2,414 validation samples with no scene overlap between the two splits. Per-source counts are given in Table 1. Waymo and nuScenes dominate the sample count, while the AVA recordings contribute a small but deliberately high-interaction subset of staged vehicle–pedestrian encounters.

Each sample is a paired sequence of ego odometry $\mathbf{e} \in \mathbb{R}^{196\times2}$ and pedestrian body motion $\mathbf{x} \in \mathbb{R}^{196\times263}$, both expressed in the pedestrian-centric frame.

Interaction scoring. Rather than hard-filtering samples, we assign each scenario a scalar *interaction score* that quantifies how strongly the pedestrian and ego vehicle interact, and use it both to bias training-time sampling and to guide cropping. Let $\mathbf{r}_t, \mathbf{g}_t \in \mathbb{R}^2$ denote the ground-plane (XZ) positions of the pedestrian pelvis and the ego vehicle at frame t. The score combines three factors:

$$s = \underbrace{\sum_t \|\mathbf{r}_{t+1} - \mathbf{r}_t\|_2}_{\text{pedestrian travel}} \cdot \underbrace{\frac{1}{T}\sum_t \mathbb{1}[\|\mathbf{g}_t - \mathbf{r}_t\|_2 < 5\,\mathrm{m}]}_{\text{proximity fraction}} \cdot \left(1 + \frac{1}{180^\circ}\sum_t |\Delta\phi_t|\right), \tag{6}$$

where $\phi_t = \mathrm{atan2}(\mathbf{r}_t - \mathbf{g}_t)$ is the pedestrian's bearing relative to the ego vehicle and $\Delta\phi_t$ is the (wrap-corrected) inter-frame change in bearing. Intuitively, a scenario scores highly when the pedestrian moves substantially, spends many frames within 5 m of the vehicle, and sweeps a large relative bearing (e.g., crossing in front of the vehicle).

Interaction-weighted sampling. During diffusion (and VAE) training we draw samples with a `WeightedRandomSampler` whose weights are the interaction scores s (floored at 0.01 so low-interaction scenarios are still occasionally seen, then normalized to sum to the dataset size). This concentrates gradient updates on the ego-relevant portion of the data distribution without discarding any samples. Evaluation uses uniform (unweighted) sampling.

Interaction-aware cropping. To bound the generation problem to a fixed horizon, we crop each scenario to $T = 196$ timesteps. For scenarios longer than 196 frames, rather than taking a random window we centre the crop on the *closest-approach frame* $t^\star = \arg\min_t \|\mathbf{g}_t - \mathbf{r}_t\|_2$, then add uniform jitter of $\pm25\%$ of the window length to the centre so the moment of closest approach is not memorized at a fixed position; the resulting window is clamped to lie within the sequence. Ego and motion streams are cropped with a shared index so they stay temporally aligned. Scenarios shorter than 196 frames are zero-padded at the end and their true length is recorded for masking.

4.2 Evaluation Protocol

We report four metrics, following the HumanML3D evaluation protocol Guo et al. [2022]:

- **FID** (Fréchet Inception Distance): computes the feature-space distance between the generated and real motion distributions, using the pretrained `t2m_moveencoder` (stride-2 Conv1D) and `t2m_motionencoder` (BiGRU) from Guo et al. [2022]. Lower is better.
- **R-Precision@1**: top-1 retrieval accuracy of the ego condition from batches of 32 candidates (1 positive + 31 negatives), using cosine similarity between the mean-pooled VAE latent (from generated motion) and the ego encoder output. Higher is better.

- **MultiModality (MM)**: average pairwise L2 distance between 10 generated samples for the same ego condition, measuring diversity per condition. Higher is better.
- **Diversity**: average pairwise L2 distance in feature space across all generated samples, measuring population-level diversity. Values *closer to the ground-truth diversity* (5.496 on our evaluation set) are better.

All reported values are the mean over **3 independent replications** with 95% confidence intervals. Unless stated otherwise, metrics are computed on the full validation split (2,414 samples), which is also used for checkpoint selection. To control for selection bias, we additionally construct a *held-out test split*: the validation scenes are partitioned into two scene-disjoint halves, one reserved purely for held-out reporting (1,190 samples from 521 scenes, never used for any model or checkpoint selection of our main system). Our best system reproduces its headline result on this held-out split (FID $= 3.34 \pm 0.19$ vs. 3.39 ± 0.18 on the full validation split), confirming the result is not an artifact of evaluating on the selection set; the conditioning-mechanism ablation (Section 5.3) is reported on this split. Note that FID and Diversity use a fixed, pretrained evaluator shared by all systems, so they are directly comparable across architectures. R-Precision, in contrast, retrieves with each model's *own* ego encoder embedding; cross-architecture R@1 comparisons (e.g., pooled vs. full-sequence encoders) should therefore be read as indicative rather than exact, as the retrieval space itself differs.

4.3 Baselines

- **H2 (Pooled, our baseline)**: EgoEncoderPooled (transformer + mean-pool $\rightarrow$ single 256D token), trans_enc denoiser (concatenated self-attention over $[\mathbf{z}, \mathbf{c}_{\text{pool}}, \mathbf{t}]$). Best at epoch 4399, CFG = 5.
- **H3 (Latent-8)**: Same as H2 but with latent dimension $L = 8$ (larger VAE). Best at epoch 4399, CFG = 5.

We additionally report four non-learned or reference baselines (Table 2) that bound the metric space. **Unconditional MLD** runs our H4 model (epoch 3399) with every ego trajectory zeroed; because CFG training dropout uses the same zeroed-trajectory construction, both guidance branches receive identical embeddings and guidance cancels exactly, yielding the model's pure unconditional prior and isolating the contribution of ego conditioning. **Retrieval** returns, for each test ego trajectory, the training motion whose ego trajectory is the nearest neighbour (32-point resampled L2). **VAE interpolation** decodes a random latent interpolation of two training motions. **Trajectory+kinematic** replaces all body features with the training mean, keeping either the ground-truth root velocity (*oracle*) or nothing (*mean*), simulating a trajectory-prediction-then-fitting pipeline with no learned body model.

4.4 Main Results

Table 3 shows the definitive evaluation results. H4 (EgoPed) achieves FID $= 3.392 \pm 0.179$ at epoch 3399 and CFG $= 10$, representing a **49% improvement** over the best baseline (H2: FID $= 6.603 \pm 0.067$ at CFG $= 5$). The improvement is significant: the confidence intervals are non-overlapping (H4 CI: $[3.213, 3.571]$; H2 CI: $[6.536, 6.670]$).

Interestingly, R-Precision@1 for H4 (0.548) is *lower* than H2's baseline (0.671). We attribute this to three compounding factors: (1) H4 generates more diverse motions per ego condition (MM: 3.956 vs. 3.503), spreading the generated distribution further in feature space and naturally reducing retrieval accuracy; (2) the R@1 metric measures alignment between generated VAE latents and ego encoder outputs — as the denoiser learns richer dynamics, this alignment metric underestimates the true conditioning fidelity; (3) R@1 retrieves with each model's own ego encoder (Section 4.2), so the pooled encoder (trained with a projection head specifically for this alignment) and the full-sequence encoder define retrieval tasks of different intrinsic difficulty. The large FID improvement confirms that H4 produces a generated distribution *much* closer to the ground truth, even as retrieval accuracy decreases.

4.5 CFG Scale Analysis

We sweep CFG $\in \{5, 10, 15\}$ for H4 at epoch 3399 (Table 4). A striking finding: H4 FID is nearly flat across CFG scales ($3.842 \rightarrow 3.617$), while H2 shows steep monotonic FID degradation at higher

Table 2: Reference baselines bounding the metric space (val split; same evaluator, normalization, and split lists as Table 3). All rows are mean ± 95% CI over 3 replications (the unconditional row resamples the diffusion model; the non-learned rows are reseeded end-to-end). Retrieval attains near-zero FID by copying real motions but is non-generative; trajectory+kinematic shows that following the root trajectory with a static body is catastrophic. Ego conditioning improves EgoPed's FID by 34% over its own unconditional prior (5.18 → 3.39) — and even the unconditional H4 prior outperforms conditionally generated H2 motions, underscoring the architectural advantage of full-sequence conditioning.

Method	FID ↓	Diversity	R@1 ↑
Retrieval (NN ego)*	0.062 ±0.004	5.34	–
Trajectory+kinematic (mean)	49.60 ±0.087	0.46	–
Trajectory+kinematic (oracle)	47.67 ±0.085	0.72	–
VAE latent interpolation	7.91 ±0.033	5.82	–
Unconditional EgoPed (ego zeroed)	5.18 ±0.209	5.96	0.031†
H2 (Pooled)	6.603	5.78	0.671
H4 (EgoPed, ours)	**3.392**	5.79	0.548

*Non-generative upper bound: copies real training motions, so low FID reflects memorization rather than conditioned synthesis. †Exactly chance level ($1/32 = 0.031$), confirming the R-Precision protocol: without ego information, retrieval is random.

Table 3: Main evaluation results. Each method is reported at its own optimal CFG (sweep in Table 4). All values are mean ± 95% CI over 3 replications. Best values in **bold**; for Diversity, values closer to the ground-truth reference are better. H4 = EgoPed (ours).

Method	Epoch	CFG	FID ↓	R@1 ↑	MM ↑	Diversity
H2 (Pooled, baseline)	4399	5	6.603 ±0.067	0.671 ±0.003	3.503	5.779
H3 (Latent-8 VAE)	4399	5	7.563 ±0.078	0.676 ±0.003	3.245 ±0.386	5.842
H4 (Ours, FREEZE_EGO=True)	3399	10	**3.392** ±0.179	0.548 ±0.002	**3.956** ±0.406	5.794
H6 (Ablation, FREEZE_EGO=False)	3399	10	5.079 ±0.034	0.352 ±0.009	2.618 ±0.309	6.064
Ground truth (reference)	—	—	—	—	—	5.496

CFG (6.603 → 7.400). This indicates that full-sequence conditioning naturally provides a rich, overdetermined conditioning signal — the denoiser attends to 196 ego tokens, so higher guidance amplifies an already high-fidelity signal rather than forcing mode collapse onto a coarse pooled token.

5 Ablation Study

5.1 Latent Dimension: 4 vs. 8 (H3)

To test whether a larger VAE latent space improves generation quality, we train a latent-8 VAE and ego-conditioned diffusion model (H3). As shown in Table 3, H3 achieves FID $= 7.563$ — **14.5% worse** than H2's 6.603 at the same epoch and CFG. R-Precision is essentially unchanged (H3: 0.676 vs. H2: 0.671), confirming that the bottleneck for conditioning quality is the ego encoder, not the latent space.

The mechanism: the fixed-capacity denoiser must now model an 8D latent distribution instead of 4D, a strictly harder task, without any additional denoiser parameters. The marginal diversity improvement (Diversity: 5.842 vs. 5.779, +1.1%) does not compensate. **Lesson**: increasing latent dimension is a poor lever for FID improvement without scaling the denoiser capacity proportionally.

5.2 Encoder Freeze vs. Unfreeze (H6)

The most important ablation is whether to freeze the ego encoder during diffusion training. H6 uses an identical architecture to H4 but with FREEZE_EGO=False, allowing the encoder to co-adapt with the denoiser via backpropagation through the diffusion loss.

Table 4: CFG sweep at epoch 3399. H4 FID is nearly flat across guidance scales; H2 degrades steeply.

Method	CFG	FID ↓	R@1 ↑	MM ↑
H2 (Pooled)	5	6.603 ±0.067	0.671 ±0.003	3.503
H2 (Pooled)	10	6.963 ±0.035	0.767 ±0.006	3.031
H2 (Pooled)	15	7.400 ±0.016	0.797 ±0.005	2.724
H4 (Ours)	5	3.842 ±0.108	0.506 ±0.009	4.141
H4 (Ours)	10	**3.392** ±0.179	0.548 ±0.002	3.956
H4 (Ours)	15	3.547 ±0.122	0.536 ±0.003	3.910

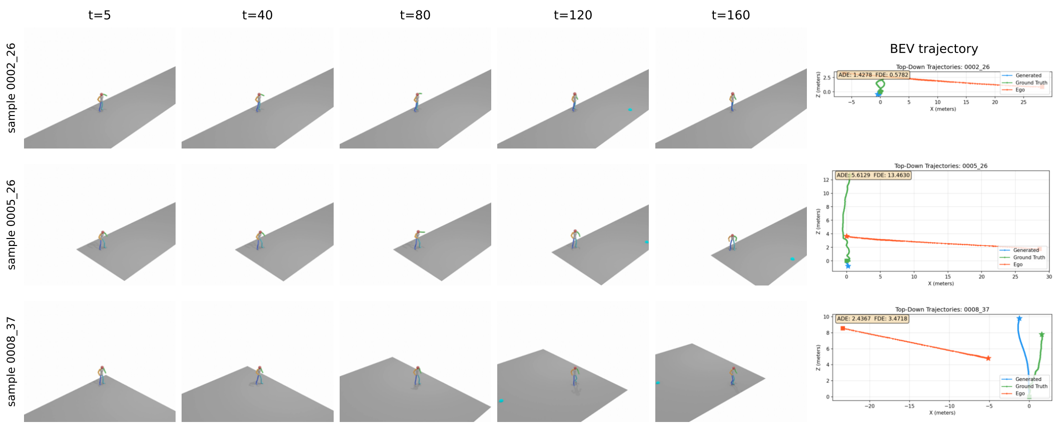

Figure 2: Qualitative results from H4 (epoch 3399, CFG=10) on three ego-conditioned samples from our test set. Each row shows five timesteps of the generated 3D body motion (HumanML3D, rendered as a rigged skeleton on a small ground plane) alongside a bird's-eye-view trajectory plot (rightmost column) showing the predicted pedestrian root trajectory (blue), ground-truth root trajectory (orange), and ego-vehicle trajectory (red). The body motion exhibits realistic gait, foot contact, and orientation changes that respond to the ego context.

Figure 3 shows the H6 training trajectory (FID and R@1 vs. epoch). Two behaviors are notable:

- **FID/R@1 tradeoff**: As training progresses, R@1 improves (encoder specializes discriminatively) but FID degrades (generation quality decreases). The two optima do *not* coincide: training-time FID is best at epoch 2099 (4.789), while R@1 continues rising long after FID has begun to deteriorate.
- **R@1 saturation far below H4**: After epoch ≈3299, training-time R@1 saturates, oscillating in the 0.37–0.42 band through epoch 4999 (maximum 0.417 at epoch 4599) while FID remains between 5.1 and 6.3. Neither metric ever approaches H4's operating point (FID 3.392, R@1 0.548) at any point in 5000 epochs.

Definitive evaluations (3 replications, CFG = 10) confirm the picture:

Table 5: H6 definitive evaluation at three checkpoints. H4 reference included for comparison.

System	FID ↓	R@1 ↑	MM ↑	Diversity
H6 epoch=2099 (best train FID)	4.952 ±0.088	0.263 ±0.009	2.455	5.957
H6 epoch=3299 (best seg-1 R@1)	5.311 ±0.060	0.348 ±0.004	2.581	6.034
H6 epoch=3399 (matched to H4 best)	5.079 ±0.034	0.352 ±0.009	2.618	6.064
H4 epoch=3399 (FREEZE=True)	**3.392** ±0.179	**0.548** ±0.002	**3.956**	5.794

H6 is worse than H4 on *all metrics at all evaluated checkpoints*. Later checkpoints cannot change this conclusion: even H6's best training-time R@1 in the saturation band (0.417 at epoch 4599,

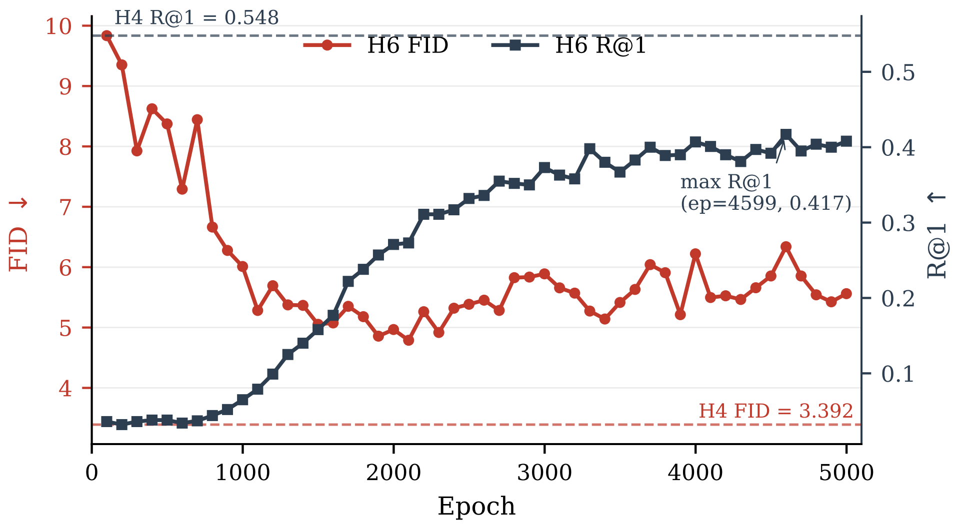

Figure 3: H6 training trajectory (FREEZE_EGO=False): training-time validation FID (left axis, red) and R@1 (right axis, blue) vs. epoch, over the full 5000-epoch run. FID is best at epoch 2099 and degrades as R@1 rises; after epoch ≈3299, R@1 saturates in the 0.37–0.42 band while FID oscillates between 5.1 and 6.3. Dashed reference lines show H4's best operating point (FID=3.392, R@1=0.548 at epoch=3399, CFG=10). H6 never approaches H4's performance on either metric.

with training-time FID 6.339) remains far below H4's definitive R@1 of 0.548 — and training-time metrics consistently *overestimate* the definitive 3-replication values (e.g., 0.398 $\rightarrow$ 0.348 at epoch 3299). The MultiModality collapse is particularly striking: H6 MM $= 2.618$ vs. H4 MM $= 3.956$ (34% lower). This indicates near-deterministic generation per condition — the ego encoder has learned highly discriminative features for each trajectory, collapsing the generated motion distribution to a narrow region.

Mechanism. When the encoder is frozen (H4), it maps ego trajectories to general-purpose representations aligned with the VAE latent space. The denoiser learns to query these fixed representations and generate diverse, plausible motions for each condition. When the encoder is unfrozen (H6), backpropagation through the denoising loss pushes the encoder toward representations that *minimize the denoising loss* — essentially learning discriminative features that uniquely identify each trajectory. This over-specialization constrains generation: the denoiser outputs similar samples for each unique ego representation, collapsing MultiModality.

This pattern mirrors the frozen-encoder paradigm in image generation Rombach et al. [2022], Ye et al. [2023], Zhang et al. [2023b]: frozen conditioning encoders preserve general-purpose representations and allow the generative model to learn diverse ways to satisfy the conditioning signal.

5.3 Conditioning Mechanism: Self-Attention vs. Cross-Attention

Our denoiser injects the ego sequence by concatenating it with the latent tokens and applying self-attention over the joint sequence (Section 4). A common alternative for sequence conditioning is *cross-attention*, where the latent tokens act as queries and the ego sequence as keys and values, so the ego tokens never attend among themselves and the attention cost is $\mathcal{O}(L \times T)$ rather than $\mathcal{O}((L+T)^2)$. To test whether the specific mechanism matters, we train an otherwise-identical cross-attention denoiser (same frozen VAE and frozen ego encoder, same data, same effective batch size) and evaluate it on the held-out test split.

As shown in Table 6, the self-attention and cross-attention denoisers achieve statistically comparable FID (3.338 vs. 3.490; 95% CIs $[3.15, 3.53]$ and $[3.43, 3.55]$ overlap), with self-attention holding a slight edge on retrieval alignment (R@1) and cross-attention generating marginally more diverse motion per condition (MM). Note that the comparison is, if anything, generous to the cross-attention variant: its best checkpoint was selected on the held-out split itself, whereas the self-attention model's checkpoint was fixed in advance from training-time validation. We also observed that the

Table 6: Conditioning-mechanism ablation. Both denoisers consume the full 196-token ego sequence and differ only in how they attend to it. Evaluated on the held-out test split (a scene-disjoint half of the validation set), CFG = 10, mean ± 95% CI over 3 replications, best checkpoint each. The two mechanisms are statistically comparable (overlapping FID intervals); cross-attention provides no advantage.

Denoiser	Attention over ego	FID ↓	R@1 ↑	MM ↑
Self-attention (EgoPed)	concatenation	**3.338** ±0.190	**0.537**	4.14
Cross-attention	queries/keys-values	3.490 ±0.059	0.501	**4.61**

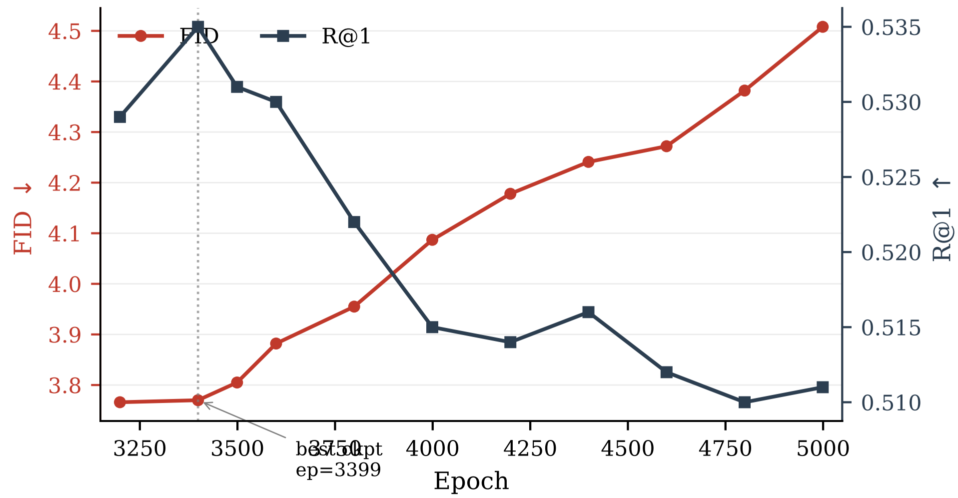

Figure 4: H4 training trajectory: training-time validation FID (left axis, red) and R@1 (right axis, blue) across the late training segment. Epoch 3399 is the best checkpoint (highest R@1, FID within noise of epoch 3199, and best definitive 3-replication FID); both metrics then degrade steadily through epoch 4999. Training loss is flat across this range — sampling quality oscillates independently of training loss.

cross-attention variant trained less stably, with a noisier FID-vs-epoch trajectory. **Takeaway**: the large improvement over pooled conditioning (Table 3) comes from exposing the denoiser to the *full* per-timestep ego sequence, not from any particular attention mechanism – either self-attention over the concatenated sequence or cross-attention works, and neither is required over the other.

5.4 Checkpoint Selection

A non-obvious finding (relevant to practitioners) is that FID does not monotonically improve with training. H2's epoch trajectory shows FID $= 6.603$ at epoch 4399, FID $= 8.400$ at epoch 4599 (regression), and FID $= 7.510$ at epoch 4999 (partial recovery) — despite flat training loss throughout. H4 similarly peaks at epoch 3399 (definitive FID $= 3.392$) and degrades steadily thereafter (Fig. 4). **Lesson**: the final checkpoint is not the best checkpoint. Periodic FID validation during training is essential; early stopping on FID is critical.

6 Discussion

Architecture recommendation. The experimental evidence strongly favors the H4 architecture: frozen ego encoder + full-sequence conditioning denoiser (self-attention over the concatenated latent and ego sequence; cross-attention works equally well, Section 5.3). Future work seeking to improve conditioning fidelity (R@1) should explore adapter-style approaches — fine-tuning a lightweight adapter on the frozen encoder while keeping the backbone fixed — rather than end-to-end unfreezing.

R-Precision@1 and the conditioning-diversity tradeoff. H4's R@1 (0.548) is lower than H2's (0.671) despite dramatically better FID. This seemingly paradoxical result reflects a genuine tradeoff: richer conditioning via the full ego sequence enables the denoiser to generate *more diverse* motions per condition, spreading the generated embedding distribution and reducing retrieval accuracy. In the context of AV simulation, diversity within valid behaviors is a feature — a single ego trajectory should plausibly lead to multiple pedestrian behaviors (crossing, waiting, accelerating). R@1 measures only conditioning alignment, not generation quality; FID captures the latter more faithfully.

Limitations.

- **Body pose quality**: Pedestrian poses are estimated (not ground-truth 3D), introducing noise in training labels. Direct 3D body annotation from datasets like JRDB-Pose or BEDLAM would improve quality.
- **Conditioning scope**: We condition only on 2D ego trajectory. Incorporating relative pedestrian-ego geometry, ego velocity profile, or map elements could improve fidelity.
- **R@1 gap**: H4 achieves $\mathrm{R@1} = 0.548$, below H2's 0.671. The adapter-style approach described above is a natural next step.
- **Physics**: Generated motions are not physically constrained (foot sliding, penetration). Post-hoc physics cleanup could improve realism for simulation deployment.
- **Evaluator domain shift**: FID and Diversity rely on motion-feature encoders pretrained on HumanML3D, whose motion distribution differs from in-the-wild pedestrian motion; absolute FID values should be compared only within our benchmark.
- **Single benchmark, no perceptual study**: All results are on our combined AVA/nuScenes/Waymo benchmark, and we do not yet report a human perceptual study of motion realism.

Broader impact. The primary intended use of ego-conditioned pedestrian motion generation is autonomous-driving simulation: synthesizing realistic, ego-aware pedestrian behaviors enables larger-scale safety testing without exposing real pedestrians to risk, and is particularly valuable for the long tail of rare or safety-critical interactions that are unsafe or impractical to collect. We note two risks. First, generated pedestrian motions could in principle be used adversarially, e.g. to seed motion-planning policies that learn to expect only nominal behaviors and consequently fail on out-of-distribution pedestrians; we mitigate this by reporting MultiModality and diversity metrics, which expose the breadth of the generated distribution rather than only its mean. Second, training data inherit demographic biases of the source AV datasets (AVA, nuScenes, Waymo): generated motion is conditioned on the distribution of pedestrians actually observed in U.S./European urban environments, so deployment in other settings should be preceded by additional validation. Generated motion data is synthetic and contains no personally identifiable information.

7 Conclusion

We introduced EgoPed, an ego-conditioned pedestrian body motion generation system based on latent diffusion with full-sequence ego conditioning. Our key finding is that exposing the denoiser to the full 196-token ego trajectory sequence dramatically outperforms single-token pooled conditioning: 49% FID improvement (6.603 $\rightarrow$ 3.392) with non-overlapping confidence intervals. A paired ablation reveals that frozen ego encoders are critical: unfreezing causes 34% MultiModality collapse as the encoder over-specializes to discriminative features.

These results establish a clear design recipe for ego-conditioned motion generation in AV settings: train the ego encoder contrastively on frozen VAE latents, then freeze it during diffusion training and condition the denoiser on the full ego sequence. This recipe produces state-of-the-art ego-conditioned full-body pedestrian motions across AVA, nuScenes, and Waymo, and provides a foundation for scalable, behavior-consistent pedestrian simulation.

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A Implementation Details

Ego encoder. 4-layer transformer encoder, 4 attention heads, 256D hidden dimension (1024D feed-forward), sinusoidal positional encoding for $T = 196$ timesteps. Input: 2D ego position sequence, projected linearly to 256D. No projection head (we verified the 2-layer MLP projection head does not improve val MSE: 1.919 vs. 1.920). Pretraining: AdamW (lr $= 1 \times 10^{-4}$, weight decay 10^{-4}), 200 epochs, batch size 64, linear warmup followed by cosine annealing, gradient clipping at max-norm 1.0.

Diffusion model. MLD denoiser, `trans_enc` architecture (self-attention over the concatenated latent, timestep, and ego tokens): 9 layers, 4 attention heads, 256D token dimension, 1024D feed-forward, latent sequence length $L = 4$. The cross-attention variant of Section 5.3 uses the same hyperparameters in a `trans_dec` configuration (latent tokens as queries, ego sequence as keys/values). Diffusion: 1000 DDPM training steps (scaled-linear β schedule), DDIM inference (50 steps, $\eta = 0$). CFG dropout rate: 10%. Optimizer: AdamW, lr $= 1 \times 10^{-4}$ (constant), 5000 total epochs. Batch size: 128. Training hardware: NVIDIA H100 (80 GB) on an academic HPC cluster; training runs in segments of up to 24h wall time.

Checkpoint selection. We select checkpoints based on validation FID (not training loss). For H4, epoch 3399 is best; we confirm this with the full training trajectory and definitive 3-replication evaluation. For H2, epoch 4399 is best despite continued training to epoch 5000.

Evaluation. All evaluations use `TEST.REPLICATION_TIMES=3` (3 independent generation passes), `DIVERSITY_TIMES=300`, `MM_NUM_SAMPLES=100`, `MM_NUM_TIMES=10`. Each replication regenerates the full test set with fresh sampling noise; the diffusion sampler is stochastic across replications while the data pipeline and evaluator are fixed (training uses seed 1234). Confidence intervals are reported as $\pm$ CI (half-width of the 95% interval across replications).

B H6 Training Trajectory Table

C NeurIPS Paper Checklist

1. **Claims.** *Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?*
   Answer: [Yes]. Justification: The two headline claims (49% FID improvement from full-sequence conditioning; MultiModality collapse from unfreezing the ego encoder) are supported by replicated evaluations with confidence intervals in Tables 3, 4, and 5.

Table 7: H6 (FREEZE_EGO=False) training-time validation metrics at selected checkpoint evaluation points (validation runs every 100 epochs; full curve in Figure 3). Training-time FID is best at epoch 2099 (4.789); R@1 rises until epoch ≈3299, then saturates in the 0.37–0.42 band (maximum 0.417 at epoch 4599) while FID remains between 5.1 and 6.3.

Epoch	Train-time FID ↓	Train-time R@1 ↑
99	9.835	0.036
499	8.373	0.038
999	6.011	0.065
1499	5.053	0.158
1999	4.967	0.271
2099	**4.789**	0.273
2499	5.385	0.332
2999	5.889	0.373
3299	5.273	0.398
3399	5.141	0.380
3499	5.415	0.367
3999	6.221	0.407
4599	6.339	**0.417**
4999	5.561	0.408

2. **Limitations.** *Does the paper discuss the limitations of the work?*
Answer: [Yes]. Justification: Section 6 lists estimated (not ground-truth) pose labels, restricted conditioning scope, the R@1 gap, lack of physical constraints, evaluator domain shift, and the absence of a perceptual study.

3. **Theory.** *Does the paper include theoretical results?*
Answer: [NA]. Justification: This is an empirical systems paper; no theoretical claims are made.

4. **Experimental reproducibility.** *Does the paper fully disclose the information needed to reproduce the main experimental results?*
Answer: [Yes]. Justification: Architecture, training, and evaluation hyperparameters are given in Section 4 and Appendix A; the dataset construction pipeline is described in Section 4.1.

5. **Code and data access.** *Does the paper provide open access to data and code?*
Answer: [No]. Justification: Code and the derived dataset will be released upon publication; the underlying nuScenes and Waymo data are publicly available under their research licenses, and the AVA recordings will be released subject to anonymization requirements.

6. **Experimental settings.** *Does the paper specify all training and test details?*
Answer: [Yes]. Justification: Optimizer, learning rate, batch size, epochs, CFG dropout, sampler settings, and checkpoint-selection protocol are in Appendix A; evaluation settings (replications, metric parameters) in Section 4.2 and Appendix A.

7. **Error bars.** *Does the paper report error bars or other statistical significance information?*
Answer: [Yes]. Justification: All key metrics report 95% confidence intervals over 3 independent generation replications; the headline FID comparison has non-overlapping intervals.

8. **Compute resources.** *Does the paper provide sufficient information on the compute resources?*
Answer: [Yes]. Justification: Single NVIDIA H100 (80 GB) per run, $\leq$24h wall-time segments, $\sim$2 segments per diffusion model (Appendix A); evaluations take $\sim$8 minutes each.

9. **Code of ethics.** *Does the research conform to the NeurIPS Code of Ethics?*
Answer: [Yes]. Justification: The work uses licensed public AV datasets and our own staged recordings with consenting participants; generated data contains no personally identifiable information.

10. **Broader impacts.** *Does the paper discuss potential positive and negative societal impacts?* Answer: [Yes]. Justification: Section 6 discusses safety benefits for AV simulation, adversarial-use and dataset-bias risks, and mitigations.

11. **Safeguards.** *Does the paper describe safeguards for responsible release?* Answer: [NA]. Justification: The model generates pedestrian body motion for simulation and poses no high misuse risk requiring gated release.

12. **Licenses.** *Are existing assets properly credited and licenses respected?* Answer: [Yes]. Justification: nuScenes and Waymo are used under their research licenses and cited; the MLD backbone, HumanML3D representation and evaluators, OmniRe, MS3D, and SMPL are cited.

13. **New assets.** *Are new assets well documented?* Answer: [Yes]. Justification: The dataset construction pipeline (annotation, pose extraction, interaction cropping) is documented in Section 4.1; documentation will accompany the release.

14. **Human subjects.** *Does the paper involve crowdsourcing or research with human subjects?* Answer: [Yes]. Justification: Staged interaction scenarios were recorded with informed, consenting participants; recordings follow institutional data-protection procedures, and released data will be anonymized.